Comparison between For Loop and While Loop in Python

Feature	For Loop	While Loop
Purpose	Used when you know how many times to repeat.	Used when you don't know how many times to repeat.
Structure	Iterates over a sequence (like a list, range, etc.).	Repeats as long as a condition is True.
Condition	Stops automatically when the sequence ends.	Stops when the condition becomes False.
Common Use	Counting, iterating through items in lists, strings.	Repeating tasks until a specific condition is met.
Example Usage	Counting from 1 to 10.	Waiting for user input or guessing a number.
Code Simplicity	Short and simple for known ranges or sequences.	Can be more flexible for dynamic conditions.
Best Use Case	Looping through items like a list, string, or range.	Looping until something happens dynamically.